HPC Day 2026 Agenda

Friday, September 18, 2026 • Hendrix Student Center

Time	Hendrix Room 121 *Floor 1*	McKissick Theater *Floor 1*	David Peebles Room *Floor 2*	Meeting Room A *Floor 2*	Meeting Room B *Floor 2*
7:30 - 8 a.m.	Registration and coffee				
8:00 - 8:30 a.m.		**Welcome** **Jon Calhoun** *Associate Professor, Electrical & Computer Engineering & Chair, ReDCAT Advisory Committee* **Jill Gemmill** *AVP, CCIT Research Computing & Data* **Brian Voss** *Vice President and CIO*			
8:30 - 8:45 a.m.		**Opening Remarks** **Provost Cole Smith**			

Time	Hendrix Room 121 *Floor 1*	McKissick Theater *Floor 1*	David Peebles Room *Floor 2*	Meeting Room A *Floor 2*	Meeting Room B *Floor 2*
8:45 - 9:15 a.m.		Keynote: **From Jazz to Supercomputing: A Calculated Journey of Happy Accidents** Barr von Oehsen *Director, Pittsburgh Supercomputing Center* A stolen guitar led to a math PhD, an ice cream social led to eighteen years at Clemson, and a chance encouragement led to the Pittsburgh Supercomputing Center - my path from jazz performance to HPC leadership was built almost entirely from unplanned pivots. This talk walks through several of them across four decades, each triggered by circumstance and redirected by curiosity. The takeaway: happy accidents compound, and the courage to walk away from a safe next step is often what makes the next one possible.			

9:15 - 10:00 a.m.		**Clemson Faculty Lightning Talks** Moderator: **Jon Calhoun** *Associate Professor, Electrical & Computer Engineering & Chair, ReDCAT Advisory Committee* Speakers: **Churamani Khanal** *Associate Professor Plant Nemotology* **AI-Driven Robotic Soil Sampler: A Need for Revolutionizing Agriculture** **Xinyi Li** *Associate Professor, Mathematical & Statistical Sciences* **Scaling Statistical Learning for Precision Health** **Siyu Huang** *Assistant Professor, School of Computing* **Build Foundations for 3D Intelligence** **Vivek Sharma** *Associate Professor, Construction and Real Estate Development* **Machine Learning Algorithm Selection Framework for Construction Industry Applications: A Conceptual Quadrant Approach**			

Time	Hendrix Room 121 *Floor 1*	McKissick Theater *Floor 1*	David Peebles Room *Floor 2*	Meeting Room A *Floor 2*	Meeting Room B *Floor 2*
		Timo Heister *Professor, School of Mathematical & Statistical Sciences* **Parallel deal.II Finite Element Simulations with deal.II: Research and Teaching** **Jerry A. Tessendorf** *Professor, School of Computing* **Man-Made Gravitational Waves: Detection and Exploitation** **Dvora Perahia** *Professor, Chemistry* **Where Computing Meets Experiment: From a Single Molecule to a Device** **Zhen Li** *Associate Professor, SMAE Mechanical Engineering* **From Static Biomedical Images to Dynamic Analysis: Personalized Hemodynamic and Biological Transport Modeling**			
10:00 - 11:00 a.m.	**Poster Session 1 and Table discussions**				

11:00 a.m. - 12:00		**Introduction to Clemson RCD LLM Service** **Doug Dawson** *Director, Cloud & Software Services, CCIT* **Carl Ehrett** *Director, Applied Machine Learning, CCIT* This session introduces the Clemson Research Computing & Data (RCD) LLM Service, a shared platform designed to provide researchers with access to open-weight models. We'll discuss the rationale behind the service, its architecture and capabilities, model selection, and research use cases.		**Where AI Meets Reality** Moderator: **Benjamin Formby** *AI/HPC Engineer Research Facilitator, CCIT* **Marc R. Birtwistle** *Professor, Chemical and Biomolecular Engineering* **Math can show sufficiency, but machine learning does it efficiently** Experimental designs for unique model inference can be suggested by using formal mathematic approaches such as linear algebra and degree of freedom analysis, but the solutions themselves are often better left for machine learning approaches rather than directly solving the mathematical formulations. Multiple examples from ongoing projects will illustrate this emerging pattern. **Aaron Masino** *Associate Professor, School of Computing* **Not all AI is agentic: Integrating biological knowledge into graph based AI for rare disease research** Developing artificial intelligence model architectures that are constrained by and utilize	**Quantum Research at Clemson** Moderator: **Jason McNeil** *Professor, Chemistry* **Justin Talbot** *Assistant Professor, Chemistry* **Understanding Chemistry with HPC: From Electrons to Molecules** High-performance computing enables quantum chemical simulations of complex molecular systems, providing molecular-level insight into electronic structure, dynamics, spectroscopy, photochemistry, and catalysis. **Enrique Saez** *Associate Professor, SMAE Mechanical Engineering* **Multiscale Materials Modeling at Palmetto** Dr. Saez will show how to use Palmetto to deploy and develop computational algorithms and tools to predict materials response. **Jorge Barraso** *Applied Computational Lecturer, Chemistry* **Quantum in the Service of Chemistry**

Time	Hendrix Room 121 *Floor 1*	McKissick Theater *Floor 1*	David Peebles Room *Floor 2*	Meeting Room A *Floor 2*	Meeting Room B *Floor 2*
				formal biological knowledge to support rare disease research and clinical decision processes. **Chris Flathmann** *Assistant Professor, School of Computing* **Creating on-premises agents for reuse across multiple human-subjects research studies.** Recent efforts to develop a reusable back-end infrastructure that allows creating a singular AI agent that can be used across multiple research studies simultaneously to reduce resource and development time.	
12:00 - 1:15 p.m.	**Lunch ;** **Table Discussions ; and** **Poster Session 2**				
1:15 - 1:45 p.m.		Keynote: **Reproducible Scientific Computing with AI Coding Agents** **D. Hudson Smith** *Assistant Professor, School of Mathematical and Statistical Sciences*			

Time	Hendrix Room 121 *Floor 1*	McKissick Theater *Floor 1*	David Peebles Room *Floor 2*	Meeting Room A *Floor 2*	Meeting Room B *Floor 2*
1:45 - 2:15 p.m.		Keynote: **From Computing to Materials** **Olga Kuksenok** *Professor, Materials Science and Engineering*			
2:15 - 2:30 p.m.	Break				

2:30 - 3:30 p.m.		Panel: **Quantum Computing Intro & Perspectives** The panel will explore how researchers can get started with quantum computing, key barriers to adoption, and its emerging applications across fields such as AI, optimization, finance, and materials science Moderator: **Zhenzhen Zhang** *Research Computing Facilitator, CCIT* Panelists: **Emily Tucker** *Assistant Professor, Industrial Engineering* **Rong Ge** *Professor, School of Computing* **Fanchen Meng** *Director Analytics, Systems & Applications, Library* **Nathan Jones** *Graduate Research Assistant, Computer Science*	Panel: **HPC in Education Roundtable**: **Integrating HPC into Academic Teaching** In this panel, Clemson faculty from diverse disciplines will discuss how they leverage Palmetto2 HPC in their classes. The discussion will identify common challenges to learning with HPC as well as the power these skills unleash in their specific disciplines. **Moderator:** **Alex Feltus** *Professor, Genetics and Biochemistry* Panelists: **Jon Cameron Calhoun** *Associate Professor, Electrical and Computer Engineering* **F. Alex Feltus** *Professor, Genetics and Biochemstry* **Timo Heister** *Professor, School of Mathematical & Statistical Sciences* **Todd Allen Price** *Lecturer, MBA Program* **Justin Talbot** *Assistant Professor, Chemistry*	**From Soft Materials to Bio Systems** Moderator: **Kimberly L. Weirich** *Assistant Professor, Materials Science and Engineering* **Feng Ding** *Professor, Physics and Astronomy* **Multiscale Modeling of Amyloidosis: From Disease Mechanisms to Mitigation Strategies.** Presenting multiscale computational approaches for uncovering the molecular mechanisms of amyloid aggregation associated with neurodegenerative diseases, including Alzheimer's disease, and how these mechanistic insights can guide the development of strategies to mitigate amyloidosis **Brian Dominy** *Associate Professor, Chemistry* **Zhen Li** *Associate Professor, SMAE Mechanical Engineering* **Multiscale modeling of biological transport and blood clotting process.** This talk will introduce how computational models	Talks: **Industry Perspectives** Moderator: **Scott Groel** *Executive Director, Research Computing and Data Infrastructure, CCIT* 2:30 – 3:00 p.m. **Cambridge Computing Chris Fowler, HPC/AI Research Scientist.** Chris focuses on computational modeling, fluid dynamics, and aligning scientific research needs with advanced IT infrastructure 3:00 – 3:30 p.m. **VAST**

Time	Hendrix Room 121 *Floor 1*	McKissick Theater *Floor 1*	David Peebles Room *Floor 2*	Meeting Room A *Floor 2*	Meeting Room B *Floor 2*
			Carlos Toxtli Hernandez *Assistant Professor, Human Centered Computing*	connect reactive transport processes across scales, reveal underlying mechanisms in biological transport and blood coagulation, and guide the development of personalized diagnostics and treatments.	
3:30 - 3:35 p.m.	Break				

Time	Hendrix Room 121 *Floor 1*	McKissick Theater *Floor 1*	David Peebles Room *Floor 2*	Meeting Room A *Floor 2*	Meeting Room B *Floor 2*
3:35 - 4:35 p.m.			Workshop: **Using Generative AI in Research with HPC** Clemson researchers discuss practical uses of generative AI for scientific code, data, modeling, and discovery on HPC systems. Moderator: **Carl Ehrett** *Director, Applied Machine Learning, CCIT* Panelists: **F. Alex Feltus** *Professor, Genetics and Biochemstry* **Rong Ge** *Professor, School of Computing* **D. Hudson Smith** *Assistant Professor, School of Mathematical and Statistical Sciences* **Carlos Toxtli Hernandez** *Assistant Professor, Human Centered Computing* **Yongkai Wu** *Assistant Professor of Electrical and Computer Engineering*	Panel Q&A: **Careers in HPC** Panelists answer questions about their career journeys Moderator: **Jill Gemmill** *AVP, CCIT Research Computing and Data* Panelists: **Rooksie Noorai** *Senior HPC Facilitator* **Jon Calhoun** *Associate Professor, Electrical & Computer Engineering & Chair, ReDCAT Advisory Committee* **Barr von Oehsen** *Director Pittsburgh Supercomputing Center* **Mike Bryant** *Director, Offensive Security* **Moises Herrera** *HPC Systems Engineer*	Talks: **Data Center Metrics / Monitoring** Moderator: **Scott Groel** *Executive Director, Research Computing and Data Infrastructure, CCIT* 3:35 – 4:05 p.m. Scott McDevitt, **Vertiv** 4:05 – 4:35 p.m. **Dell + Intel**

Time	Hendrix Room 121 *Floor 1*	McKissick Theater *Floor 1*	David Peebles Room *Floor 2*	Meeting Room A *Floor 2*	Meeting Room B *Floor 2*
4:40 - 5:15 p.m.		**Palmetto 2 Town Hall** The User community is invited to ask questions, comment on Palmetto2 services and offer suggestions **Closing Remarks and Student Poster Awards** **Jon Calhoun** *Associate Professor, Electrical & Computer Engineering & Chair, ReDCAT Advisory Committee*			